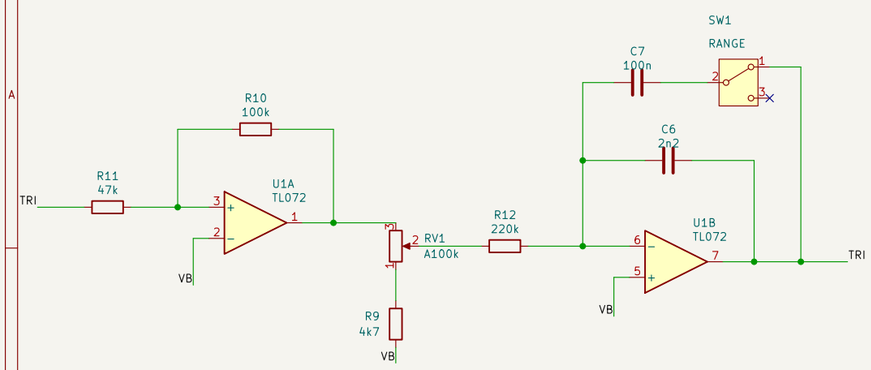
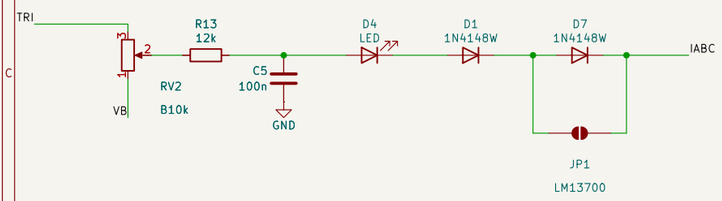
LFO: SCHMITT (U1A) + INTEGRATOR (U1B), RATE, RANGE switch



LFO ~1–24 Hz with C7 switched in; ~50 Hz – 1.1 kHz on C6 alone (46x).

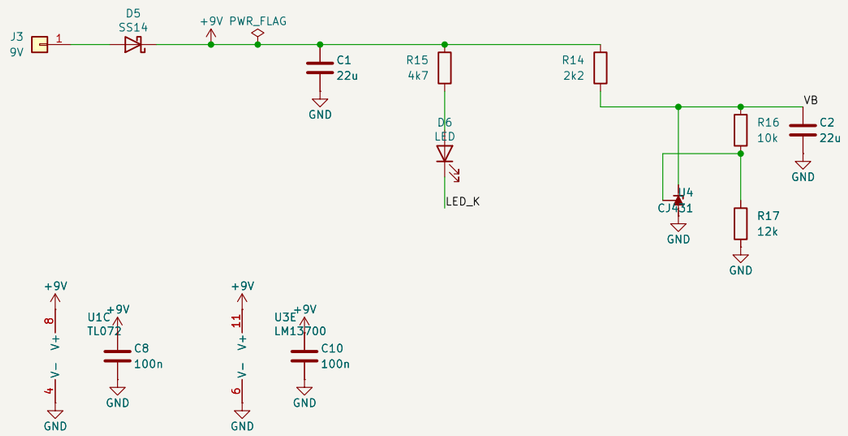
DEPTH -> OTA CONTROL CURRENT (rate LED D4)



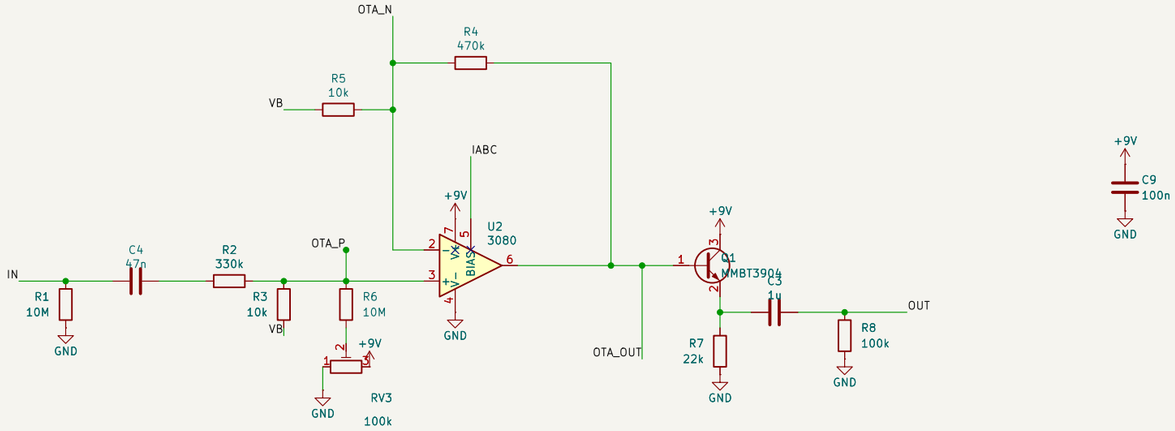
I_ABC ~ (DEPTH wiper – 3.6 V) / 12k: 0 to ~200 uA at full DEPTH, ~80 uA at DEPTH minimum.

3080: JP1 open. LM13700: bridge JP1.

POWER, VB = 4.5 V (TL431), LEDs & PADS



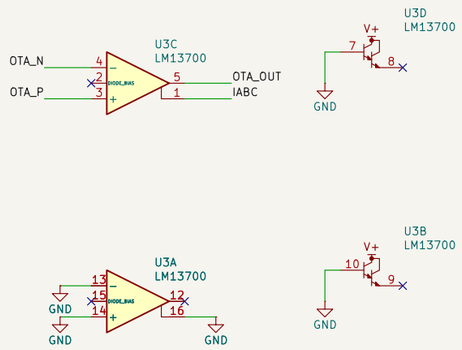
VCA: 3080 OTA in DIP-8 SOCKET (U2) + Q1 OUTPUT FOLLOWER



TRIM (through R6 10M) nulls the LFO bleed-through at the OTA input – most important in AUDIO range.

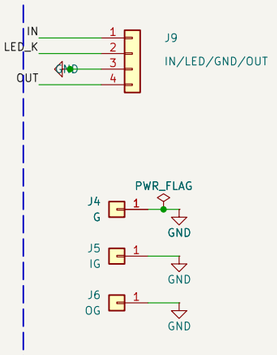
U2 = 3080 (socket) OR U3 = LM13700 (SOIC-16, below): same pins via OTA_P / OTA_N / OTA_OUT / IABC.

ALTERNATIVE OTA: LM13700 half A (U3) – fit INSTEAD of U2, used as a plain 3080



Unused: half B inputs + I_ABC to GND, both buffers off, diode-bias pins open.

OFF-BOARD WIRING



DRAFT for review
TL072 LFO: RATE, DEPTH, RANGE (LFO / audio-rate pseudo ring mod), rate LED
After John Hollis. 3080 in a socket OR LM13700 (fit one); TL431 4.5 V bias
Super-Freq
Sheet: /
File: Frobnicator.kicad_sch

Title: Frobnicator SMD

Size: A3	Date: 2026-09-29	**Rev: 1.0 draft**
KiCad E.D.A. 10.0.4		Id: 1/1